

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

– Ho Chi Minh, October 25 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
25/10/2026	A	Nguyễn Trần Minh Hưng	Added information and diagrams for section II; 1, 2, 5
25/10/2026	A	Đinh Hoàng Kha	Added information and diagrams for section III; 3.1

26/10/206	M	Đinh Hoàng Kha	Complete section III; 3.1
26/10/2026	M	Nguyễn Trần Minh Hưng	Complete section II 1, 2, 5

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Product Overview

The Intelligent Hotel Concierge Robot (HCRobot) is a software system for a robot-assisted concierge service. It supports guests seeking hotel information, service recommendations, and assistance, while enabling hotel staff to handle requests that require human intervention. Its purpose is to improve guest service consistency and reduce staff workload, particularly when several inquiries arrive at the same time.

HCRobot combines multilingual conversational interaction with guest service request handling, robot-facing interaction functions, staff collaboration, and administrative management. Guests can interact through voice or a display; the proposal also identifies an Interaction Screen for Guest, a Robot Application, a Mobile App for Hotel Staff, and a Web App for Admin.

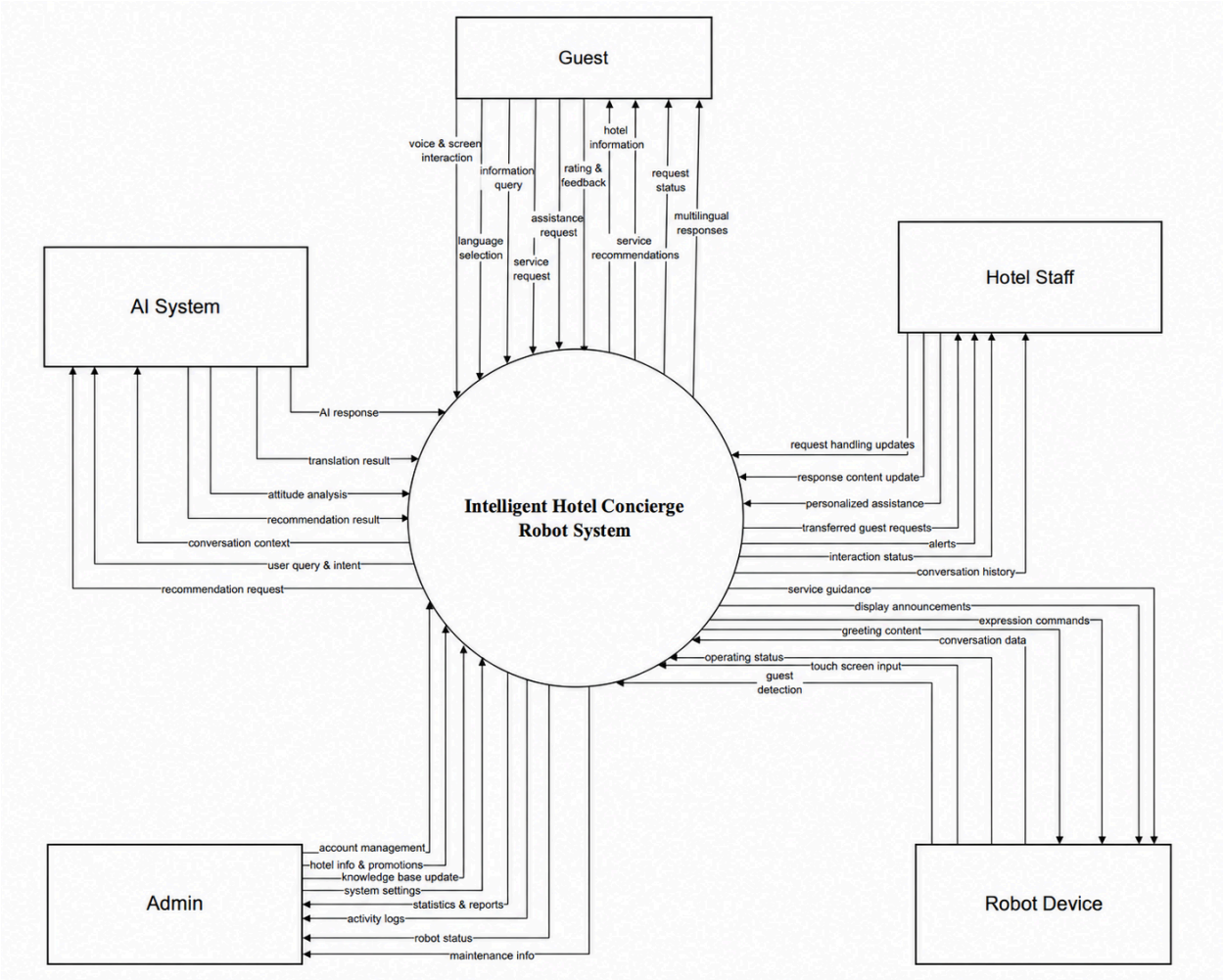


Figure 1. HCRobot context diagram

2. User Requirements

2.1 Actors

Table 2 identifies the external participants in the HCRobot boundary. Actor IDs are used consistently in the use-case descriptions. A module is not automatically an actor; in particular, the Robot application is internal and Robot Device is external.

Table 2. Actor descriptions

ID	Actor	Description
1	Guest	Primary human actor. Uses voice or screen interaction to select a language, obtain information and recommendations, submit and track service requests, request staff assistance, and rate the interaction. A guest account or room-verification requirement is not specified.

2	Hotel Staff	Primary human actor. Handles transferred requests, updates their status, provides personalized assistance, monitors interactions and robot status, receives alerts, reviews records, and updates response content and knowledge.
3	Admin	Primary human actor. Manages accounts and permissions, hotel content, the knowledge base, robot records and maintenance schedules, statistics, logs, reports, and settings.
4	AI System	Supporting external software actor. Exchanges conversational inputs and context, AI responses, translations, and recommendation results. S2 also shows attitude analysis; its interpretation and permitted use require confirmation.
5	Robot Device	External hardware actor and event source. Supplies guest-detection events, touch/screen input, operating status, and conversation data; receives greeting content, expression commands, announcements, and service guidance. The Robot application remains inside the software boundary.

2.2 Use Cases

2.2.1 Diagram(s)

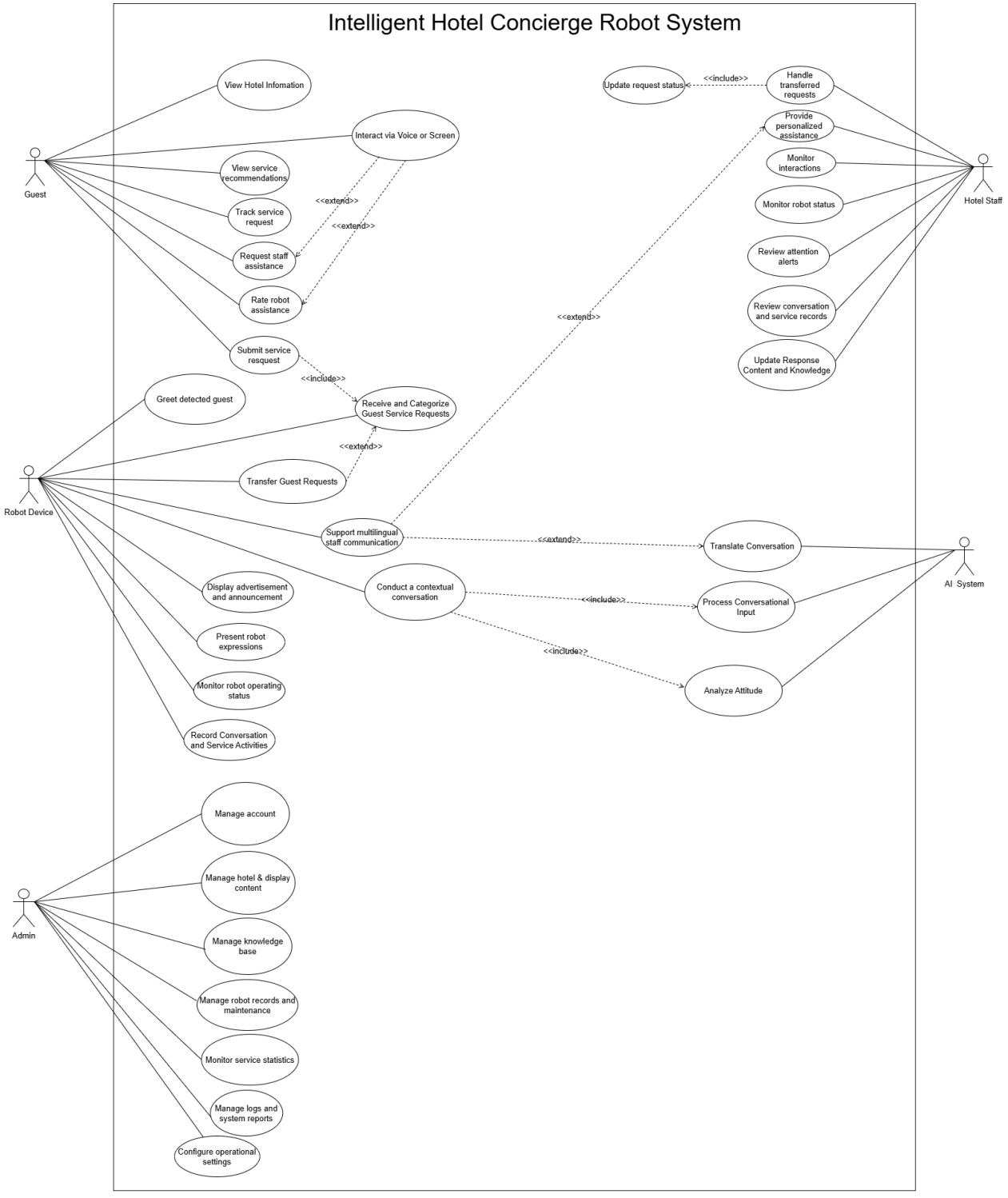


Figure 2. HCRobot use case diagram

2.2.2 Descriptions

ID	Use Case	Actors	Use Case Description
01	View Hotel Information	Guest	The guest views information about hotel facilities, amenities, promotions, policies, and nearby attractions.
02	Interact via Voice or Screen	Guest	The guest communicates with the robot through voice commands or the display and selects a preferred interaction language.
03	View Service Recommendations	Guest	The guest receives recommendations for hotel services, dining, transportation, and local points of interest.
04	Track Service Request	Guest	The guest checks the current status of a previously submitted service request.
05	Request Staff Assistance	Guest	The guest requests help from hotel staff when robot assistance is insufficient or human intervention is needed.
06	Rate Robot Assistance	Guest	The guest submits a rating or feedback after interacting with the robot.
07	Submit Service Request	Guest	The guest submits a request for housekeeping, extra amenities, wake-up calls, or transportation assistance.
08	Greet Detected Guest	Robot Device	The system receives a guest-detection event and presents an appropriate greeting through the robot.
09	Receive and Categorize Guest Service Requests	Robot Device	The system receives the guest request and categorizes it according to the requested service.
10	Transfer Guest Requests	Robot Device	The system forwards requests requiring human intervention to the appropriate hotel staff.
11	Support Multilingual Staff Communication	Robot Device, AI System	The system supports translation during communication between international guests and hotel staff.
12	Display Advertisement and Announcement	Robot Device	The robot displays hotel information, announcements, and promotions while idle.
13	Conduct a Contextual Conversation	Robot Device, AI System	The system maintains conversation context and provides relevant multilingual responses to guests.
14	Present Robot Expressions	Robot Device	The robot presents simple expressions and responses during interactions with guests.
15	Monitor Robot Operating Status	Robot Device	The system monitors robot operation and identifies abnormal conditions that require attention.
16	Record Conversation and Service Activities	Robot Device	The system records conversation history and guest service activities for authorized review.
17	Handle Transferred Requests	Hotel Staff	Hotel staff receive and handle guest service requests transferred by the robot.
18	Update Request Status	Hotel Staff	Hotel staff update service request progress so guests can track the latest status.
19	Provide Personalized Assistance	Hotel Staff	Hotel staff provide assistance when a guest request requires human attention.

20	Monitor Interactions	Hotel Staff	Hotel staff monitor ongoing guest interactions with the robot.
21	Monitor Robot Status	Hotel Staff	Hotel staff view robot operating status for service supervision.
22	Review Attention Alerts	Hotel Staff	Hotel staff view alerts relating to urgent guest requests or abnormal robot conditions.
23	Review Conversation and Service Records	Hotel Staff	Hotel staff review conversation history and guest service records to support request handling.
24	Update Response Content and Knowledge	Hotel Staff	Hotel staff update response content and service knowledge to improve assistance quality.
25	Manage Account	Admin	The administrator manages staff accounts and access permissions.
26	Manage Hotel and Display Content	Admin	The administrator manages hotel information, amenities, promotions, and content displayed by the robot.
27	Manage Knowledge Base	Admin	The administrator manages hotel knowledge and guest service information used by the system.
28	Manage Robot Records and Maintenance	Admin	The administrator manages robot profiles, operating records, and maintenance schedules.
29	Monitor Service Statistics	Admin	The administrator monitors guest interactions, service requests, and satisfaction statistics.
30	Manage Logs and System Reports	Admin	The administrator reviews activity logs, conversation records, and system reports.
31	Configure Operational Settings	Admin	The administrator configures system settings and operational parameters.
32	Translate Conversation	AI System	The AI service provides translated content to support multilingual guest–staff communication.
33	Process Conversational Input	AI System	The AI service processes conversational input and supplies responses relevant to the interaction context.
34	Analyze Attitude	AI System	The AI service analyzes guest attitude, including facial-expression information when available, and returns an analysis result to the system.

3. Functional Requirements

3.1 System Functional Overview

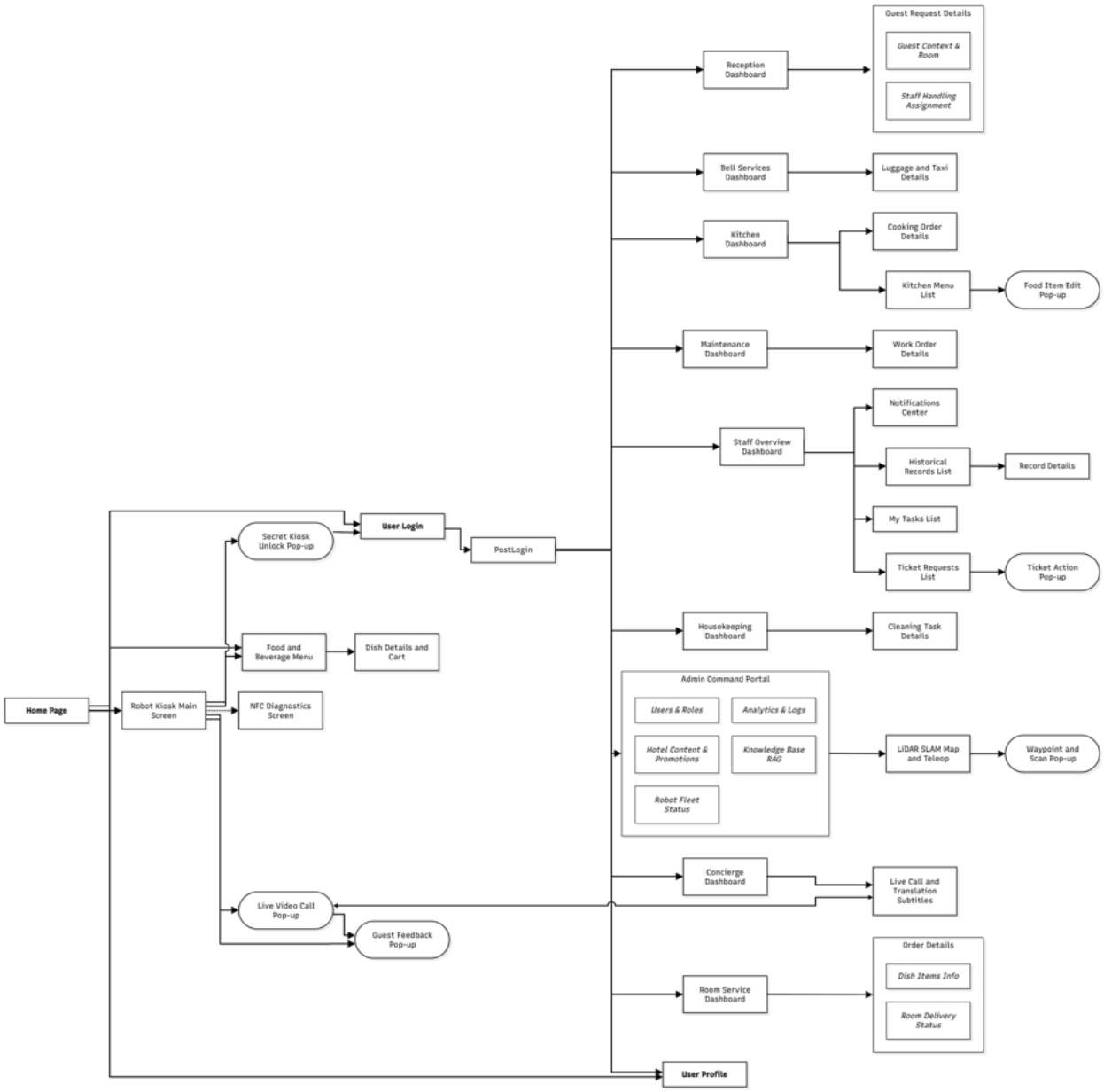
The Intelligent Hotel Concierge Robot (HCRobot) software system operates across four primary interactive subsystems:

1. **Robot Touchpoint & Edge Client**: Responsive full-screen PWA (progressive web app) interface mounted directly on the robot chassis (displaying animated avatar expressions, voice wave visualizer, multilingual concierge prompts, interactive food and beverage menu, wayfinding navigation map, and WebRTC live call overlay).

2. **Hotel Staff Operations Console**: Department-specialized web dashboards providing real-time ticket triage, order processing, task checklists, live calling escalation with real-time translation subtitles, and push notification feeds.
3. **Admin Command Portal & Teleop Console**: Administrative web application providing comprehensive account governance, hotel content and idle promotions management, Obsidian/ChromaDB RAG knowledge synchronization, robot fleet telemetry, and real-time LiDAR 2D SLAM mapping and teleoperation.
4. **Central Backend & Robotics Bridge**: High-performance FastAPI asynchronous server managing database persistence (PostgreSQL via SQLAlchemy 2.0 Async), ChromaDB vector search, AI speech services (Faster-Whisper STT, EdgeTTS), real-time Socket.IO event gateway, WebRTC P2P signaling, and ROS 2 robotics telemetry on Raspberry Pi 5.

3.1.1 Screens Flow

Diagram Viewer



3.1.2 Screen Descriptions

#	Feature	Screen	Description
1	User Authentication & Profile	Home Page	Public entry point for system features and role navigation.
2	User Authentication & Profile	User Login	Authentication interface for staff and administrators.
3	User Authentication & Profile	User Register	Registration screen for new internal accounts.
4	User Authentication & Profile	Reset Password	Password recovery request screen.

5	User Authentication & Profile	Change Password	Self-service password modification for authenticated users.
6	User Authentication & Profile	User Profile	Displays and edits current user details and preferences.
7	Guest Touchpoint	Robot Idle & Promotion Display	Displays idle announcements, animations, and promotional banners when no guest is interacting.
8	Guest Touchpoint	Robot Voice & Touch Interface	Main guest interface featuring speech visualizer, multilingual prompts, and category selection.
9	Guest Touchpoint	Hotel Facility & Amenity Info	Provides details and operating hours for hotel amenities and services.
10	Guest Touchpoint	Food & Beverage Menu & Ordering	Interactive ordering screen for in-house dining and room service items.
11	Guest Touchpoint	Local Attractions & Recommendations	Curated guide to nearby points of interest and activities.
12	Guest Touchpoint	Wayfinding & Indoor Navigation Map	Interactive visual floor maps and directions to key hotel areas.
13	Guest Touchpoint	Service Request Submission	Form interface to submit requests for housekeeping, amenities, or assistance.
14	Guest Touchpoint	Service Request Status Tracking	Screen allowing guests to check progress of active service requests.
15	Guest Touchpoint	WebRTC Staff Call Overlay Pop-up	Pop-up modal facilitating direct video/audio calling with hotel staff.
16	Guest Touchpoint	Guest Rating & Feedback Pop-up	Feedback pop-up displayed at the conclusion of an interaction.
17	Hotel Staff Operations Console	Staff Operations Dashboard	Central hub for staff to review pending tasks, alerts, and system status.
18	Hotel Staff Operations Console	Service Request Triage Board	Kanban/list view of incoming guest requests for assignment and tracking.
19	Hotel Staff Operations Console	Request Detail & Status Update	Detailed view to update status, add resolution notes, or reassign tickets.
20	Hotel Staff Operations Console	F&B Kitchen Order Processing Queue	Kitchen display queue for managing meal preparation and order fulfillment.
21	Hotel Staff Operations Console	WebRTC Live Call Console	Staff calling interface featuring real-time translated subtitles.
22	Hotel Staff Operations Console	Conversation & Service History Review	Searchable archive of past guest interactions and request logs.
23	Hotel Staff Operations Console	Quick Response Content Management	Allows staff to edit canned responses and rapid assistance templates.
24	Hotel Staff Operations Console	Attention Alerts Feed Pop-up	Real-time alert pop-up for high-priority events or robot fault notifications.
25	Admin Command Portal & Teleop	Admin Analytics & Control Portal	High-level executive dashboard showing system metrics and operational health.
26	Admin Command Portal & Teleop	User Account & Role Governance	User management table for account creation, role assignment, and access rights.
27	Admin Command Portal & Teleop	User Details & Permission Matrix	Granular authorization setting screen for individual user profiles.

28	Admin Command Portal & Teleop	Hotel Content & Idle Promotion Manager	CRUD interface for hotel amenities, promotions, and idle displays.
29	Admin Command Portal & Teleop	RAG Knowledge Base Sync (Obsidian/ChromaDB)	Interface to synchronize knowledge markdown files with ChromaDB vector storage.
30	Admin Command Portal & Teleop	Knowledge Document Editor	In-app editor for updating AI knowledge articles and guidelines.
31	Admin Command Portal & Teleop	Robot Fleet Manager	Fleet status matrix displaying telemetry, battery level, and maintenance state.
32	Admin Command Portal & Teleop	Robot Profile & Maintenance Schedule	Detailed profile page for configuring robot hardware profiles and service logs.
33	Admin Command Portal & Teleop	Service Performance & Rating Reports	Reporting page for guest satisfaction scores and service metrics.
34	Admin Command Portal & Teleop	Audit Logs & Export Manager	System audit trail viewer with log export capabilities.
35	Admin Command Portal & Teleop	System Configuration & Thresholds	Global settings for application parameters, thresholds, and integrations.
36	Admin Command Portal & Teleop	F&B Menu Admin & Category Manager	Administrative setup page for dining menus, categories, and item pricing.
37	Admin Command Portal & Teleop	Robot LiDAR SLAM & Teleoperation	Real-time LiDAR 2D SLAM mapping dashboard and manual teleoperation controls.
38	Admin Command Portal & Teleop	Waypoint and Scan Pop-up	Pop-up window for setting navigation waypoints and managing LiDAR scan triggers.

3.1.3 Screen Authorization

Screen / Activity	Guest	Front Desk	House-keeping	Bell Service	Maintenance	Kitchen	Concierge	Admin
Home Page	X	X	X	X	X	X	X	X
User Login		X	X	X	X	X	X	X
User Profile		X	X	X	X	X	X	X
Robot Screen	X							
Staff Dashboard & Alerts		X	X	X	X	X	X	X
Service Request Triage & Status		X	X	X	X		X	X
F&B Kitchen Order Queue						X		X
WebRTC Live Call Console		X					X	X
History Review & Quick Content		X	X	X	X	X	X	X
Admin Portal & Analytics								X
User Governance & Matrix								X
Content, RAG & F&B Admin								X

Robot Control & Maintenance					X			X
Reports, Audit Logs & Config								X
LiDAR SLAM & Teleoperation					X			X

3.1.4 Non-Screen Functions

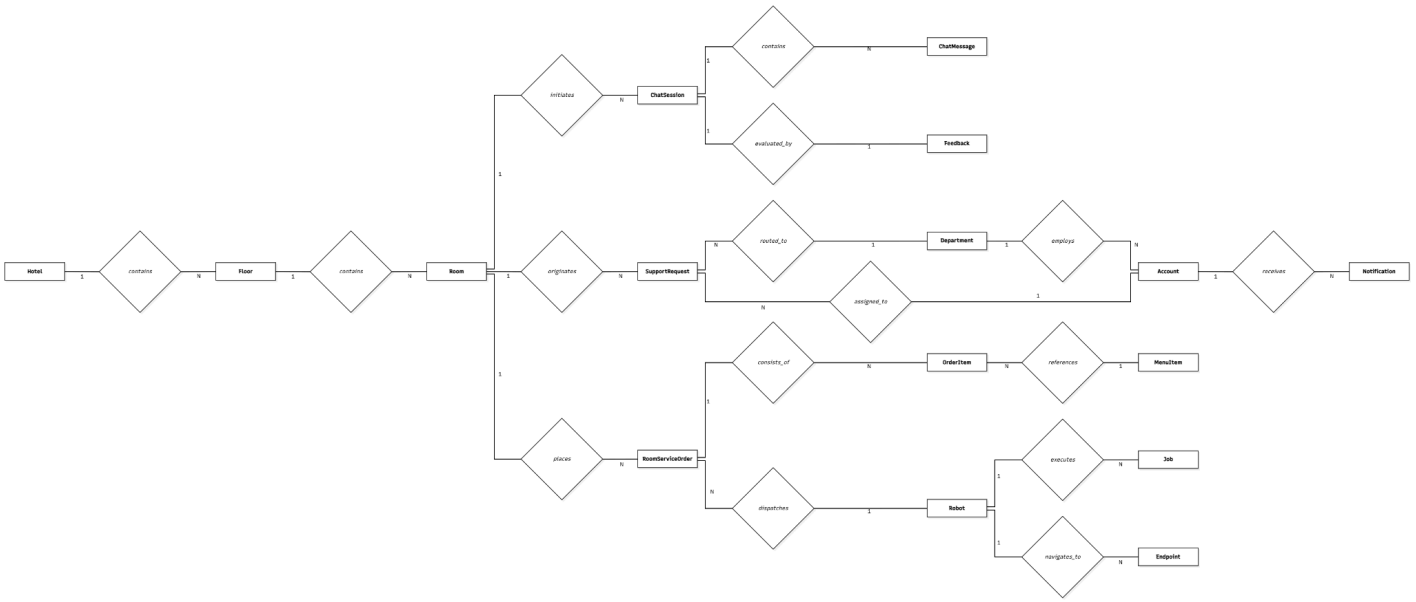
The following table details the non-screen system functions, including background services, robotics daemons, AI processing pipelines, and API integrations within the HCRobot system.

#	Feature	System Function	Description
1	AI Speech Engine	Faster-Whisper STT Pipeline	Asynchronous audio processing pipeline triggered upon voice audio buffers received from the robot microphone. Performs voice activity detection (VAD), audio normalization, and decodes speech into multilingual text transcripts with word-level timestamps and confidence metrics.
2	AI Speech Engine	EdgeTTS Voice Synthesis Service	Asynchronous streaming service triggered upon conversational AI responses or system announcements. Synthesizes natural-sounding neural speech audio, delivers chunked audio to robot speakers, and generates synchronized viseme events for facial lip-sync animation.
3	Knowledge Base & RAG	ChromaDB Vector Retrieval Engine	Semantic retrieval service triggered on guest queries or admin knowledge synchronization. Parses hotel policy and service markdown documents into semantic chunks, generates dense vector embeddings via Sentence Transformers, and conducts cosine similarity retrieval to ground LLM responses with accurate hotel context.
4	Conversational AI	LLM Intent Extraction & Function Calling	Natural language understanding service triggered on processed guest text input. Evaluates dialogue turns using local Ollama (Llama 3 / Qwen) or cloud LLM, extracts structured JSON entities (e.g., room number, requested amenity, pickup time), and dispatches automated service ticket creation calls without manual form entry.
5	Robotics & Mapping	RPLIDAR 2D SLAM Mapping Daemon	Continuous background daemon active during mapping mode via slam_toolbox/Cartographer. Ingests 360-degree laser scan distance arrays from RPLIDAR A1/A2 over USB serial, estimates odometry pose, computes 2D occupancy grid updates, and serializes .yaml and .pgm map files to disk.
6	Autonomous Navigation	Nav2 Path Planner &	Background robotics service triggered upon destination dispatch or automated patrol schedules. Calculates global costmap paths (NavFn/Smac) and collision-free local

		Controller Daemon	trajectories (DWB controller), sending /cmd_vel motor speed commands to the ESP32 differential driver while avoiding static and dynamic obstacles.
7	Hardware Safety	Ultrasonic & Bumper Collision E-Stop Daemon	Real-time safety daemon executing in a 50Hz continuous control loop on the Robot Edge Node / ESP32. Ingests multi-directional HC-SR04 ultrasonic distance sensors; immediately publishes emergency stop (E-STOP) override and halts drive motors if any obstacle enters the critical safety envelope (< 25 cm).
8	Real-Time Communication	Socket.IO Event Broadcasting Service	Persistent bidirectional WebSocket gateway triggered upon system state changes (ticket creation, status update, call escalation, robot heartbeat). Broadcasts real-time JSON payloads across connected staff browsers, admin consoles, and robot to ensure zero-latency UI synchronization.
9	Media & Escalation	WebRTC P2P Signaling Gateway	Real-time media signaling service triggered on human concierge escalation calls. Manages SDP offer/answer exchanges and ICE candidate gathering between robot camera/mic and concierge browser, establishing low-latency, encrypted peer-to-peer HD video/audio streams.
10	System Health	Robot Telemetry & Heartbeat Monitor	Recurring background cron worker executing every 5 seconds. Ingests robot telemetry heartbeats (battery state of charge, IP address, ROS 2 node health, motor driver status) and automatically flags the robot entity as OFFLINE or WARNING if heartbeats lapse for over 15 seconds.
11	Governance & Security	Audit Logging & Route Interceptor	Synchronous HTTP middleware executing on every REST API request. Intercepts incoming and outgoing HTTP traffic via FastAPI RouteInterceptor, recording caller identity, IP address, HTTP method, target entity ID, status code, and audit payloads into the PostgreSQL database.
12	Idle Mode & Promotion	Scheduled Idle Advertisement Scheduler	Background cron timer evaluating robot inactivity every 60 seconds. Automatically transitions the display to hotel promotional slideshow banners and triggers periodic multilingual greeting announcements during configured daytime operational windows.

3.1.5 Entity Relationship Diagram

Diagram Viewer



Entities Description

#	Entity	Description
1	Hotel	Hotel property profile and general information configuration.
2	Floor	Building level classification for navigation, room assignment, and mapping.
3	Room	Guest room details, location coordinates, and operational state.
4	ChatSession	Interactive conversational interaction session between guest and robot.
5	ChatMessage	Individual text or transcript turn within a chat session.
6	Feedback	Guest rating score and qualitative comments on robot assistance.
7	SupportRequest	Guest-submitted ticket for assistance, housekeeping, or amenities.
8	Department	Organizational division within the hotel responsible for specific service requests.
9	Account	System account credentials and profile details for hotel staff and administrators.
10	Notification	Real-time alert payload delivered to staff devices or admin console.
11	RoomServiceOrder	Food and beverage room service order transaction.
12	OrderItem	Individual menu item entry within a room service order.
13	MenuItem	Food or beverage item specification, pricing, and description.
14	Robot	Concierge robot unit hardware profile, battery status, telemetry, and operational state.

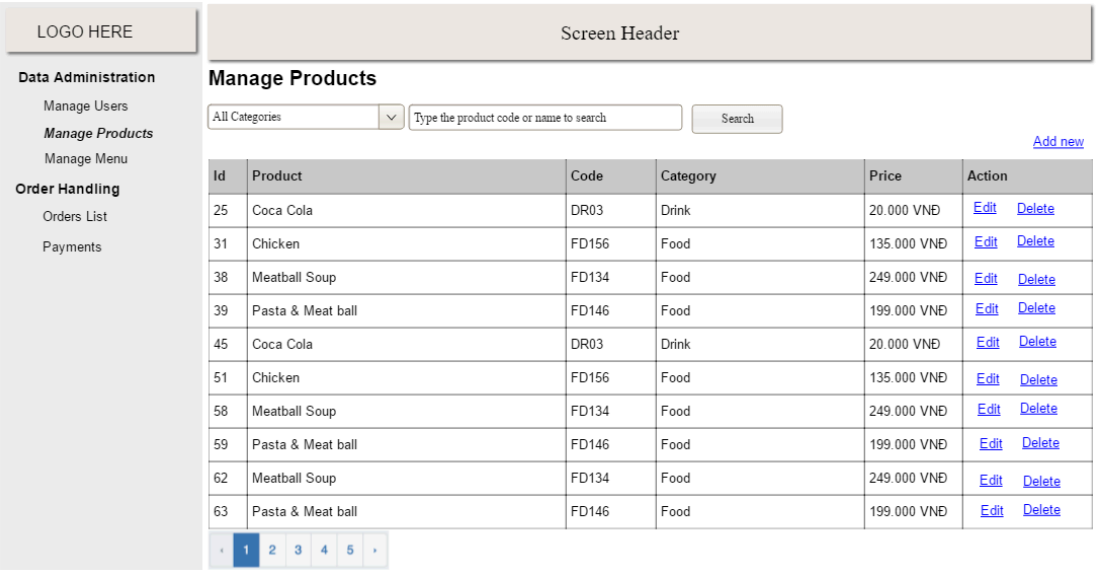
15	Job	Scheduled maintenance or automated tasks assigned to a robot unit.
16	Endpoint	Target location waypoint for autonomous robot navigation dispatch.

3.2 <<Feature Name 1>>

3.2.1 <<Function Name 1>>

[A function can be a screen or a non-screen function (listed in the part 3.1.5 above). In this part, you need to provide the details on the related function, focus on mentioning below information

- *Function trigger: how this function is triggered (navigation path, a timing frequency, etc.*
- *Function description: actors/roles, purpose, interface, data processing, etc.*
- *Screen layout: mock-up prototype of the screen, sample below is for Manage Products screen*



- *Function Details: provide explanation for the data, validation, business rules, functionalities (for both normal cases and abnormal cases), etc. of the function so that the reader can image how it work.]*

3.2 Authentication & Profile

This feature controls who can enter the HCRobot web/PWA console and which workspace each account lands on. It covers the Account entity.

3.2.1 Login

Function trigger

- User opens the system URL and clicks **Login** on the Landing page, or navigates directly to `/login`.
- Any protected route (`/staff/*`, `/admin/*`) opened without a valid session redirects to `/login`.

Function description

- Actors: Admin, Manager, Hotel Staff (all departments), Robot device account.
- Purpose: authenticate the user and route them to the correct workspace based on role and department.
- Interface: a login form with Username, Password (show/hide toggle), Login button and Back to Home link.
- Data processing: the frontend sends the credentials to `POST /api/v1/auth/login`. The server verifies the password hash, issues a JWT access token (HS256, valid 7 days) and returns the user profile plus a target dashboard. The client stores the token and redirects.

Function details

Field	Type	Required	Validation
Username	Text	Yes	Trimmed, case-insensitive. Department aliases are accepted (e.g. "frontdesk" → reception, "bep" → kitchen).
Password	Password	Yes	Masked by default; eye icon toggles visibility.

Business rules:

1. BR-AUTH-01: Both fields are required. If either is empty, the screen shows "Please enter both username and password" and no request is sent.
2. BR-AUTH-02: After successful login the user is routed by role/department: Admin → Admin Portal (`/admin/portal`); Robot account → Robot Screen (`/robot`); Staff → own department dashboard (Reception, Housekeeping, Bell Services, Maintenance, Kitchen, Room Service, Concierge). Unknown department defaults to Reception.
3. BR-AUTH-03: Every successful login is written to the audit log (action LOGIN, actor, role, department).
4. BR-AUTH-04: A user who is already logged in and opens `/login` is redirected straight to their workspace.

Abnormal cases:

- Wrong username or password → HTTP 401; red banner "Username or password is incorrect". A LOGIN_FAILED warning is logged with the attempted username.
- Account deactivated or soft-deleted (`is_active = false` or status Inactive/Deleted) → HTTP 403; message "This account has been disabled. Please contact the Administrator."
- Server unreachable → generic message "Login failed. Please check again."; the form keeps the entered username.

3.2.2 Logout and Kiosk Secure Exit

Function trigger

- Staff/Admin: click the avatar menu in the header → **Logout**.

- Robot screen (kiosk mode): no visible logout button. Staff trigger a hidden gesture: tap the HCROBOT logo or the invisible 64×64 px bottom-right corner 5 times within 2 seconds, long-press the bottom-right corner for 3 seconds, press `Ctrl + Shift + L`, or press `Esc` 3 times within 1.5 seconds.

Function description

- Actors: Admin, Hotel Staff, Robot device account (operated by staff/technician).
- Purpose: end the session safely, and prevent guests or children from closing the robot application.
- Interface: on the robot screen, a "Robot Logout Password" modal with one password field, Confirm and Cancel buttons.
- Data processing: the client calls `POST /api/v1/auth/logout`, which records a LOGOUT audit entry, then clears the stored token and returns to the Landing page.

Function details

1. BR-AUTH-05: The logout API always responds successfully so the client can clear its local session even if the token is already expired.
2. BR-AUTH-06: On the robot screen, logout requires a valid kiosk exit password (one of the authorized maintenance passwords). The robot stays locked on the guest screen until the correct password is entered.
3. BR-AUTH-07: The tap counter resets if the 5 taps are not completed within 2 seconds.

Abnormal cases:

- Wrong kiosk password → error "Incorrect password"; the modal stays open and the robot screen remains active.
- Cancel or tap outside the modal → modal closes, no logout.
- Network error when calling the logout API → local session is still cleared; the audit entry is not written.

3.2.3 View and Update Profile / Change Password

Function trigger

- Staff: sidebar → **Profile** (`/staff/profile`).
- Admin: avatar menu → **My Profile** (User Profile modal).

Function description

- Actors: Admin, Manager, Hotel Staff.
- Purpose: let users keep their personal information and password up to date.
- Interface: profile card (avatar, full name, staff code, role, department, location, status) with an Edit form, and a separate Change Password panel (Current, New, Confirm password, each with show/hide).
- Data processing: profile changes call `PATCH /api/v1/auth/profile`; password changes call `POST /api/v1/auth/change-password`. The new password is hashed before it is stored.

Function details

Field	Editable	Validation
Avatar	Yes	Upload PNG, JPG, WEBP or GIF, max 5 MB; or pick a preset avatar; can be removed.
Full name	Yes	Required, cannot be blank.
Location, Status	Yes	Shift status values: Available, Busy, Off shift.
Username, Staff code, Role, Department	No (read-only for staff)	Changed only by Admin in Account Management.
Current password	–	Required.
New password	–	At least 6 characters.
Confirm password	–	Must match New password.

Business rules:

1. BR-AUTH-08: An uploaded avatar is only saved when the user clicks **Save changes**.
2. BR-AUTH-09: Password change requires the correct current password.

Abnormal cases:

- Invalid image type or file over 5 MB → "Please choose a valid image (PNG, JPG, WEBP, GIF)" / "Maximum image size is 5 MB".
- Blank full name → "Please enter full name".
- New password under 6 characters, or confirmation mismatch → validation message; nothing is sent.
- Wrong current password → HTTP 400 "Current password is incorrect".
- Expired token → HTTP 401; user is returned to the Login screen.

3.3 Robot Guest Interaction

This feature is the guest-facing kiosk running on the robot's touchscreen as a PWA (`/robot`, also installable on a phone). Guests never log in; the robot device is signed in with a Robot account. Entities: ChatSession, ChatMessage, SupportRequest, RoomServiceOrder, OrderItem, MenuItem, Feedback, Notification.

The robot screen works as a state machine:

State	Name	Meaning
RT-01	Sleeping	No guest in front of the robot; light idle face.
RT-02	Welcome	Guest detected; robot greets and offers help.
RT-03	Listening	Microphone is open and the transcript is shown live.
RT-04	Processing	Question sent to the AI engine; thinking animation.
RT-05	Route Guidance	Robot shows the floor map, route and QR code to a destination.
Standby	Advertisement	Full-screen promotions after the robot stays idle in RT-01.

3.3.1 Guest Detection and Greeting

Function trigger

- Automatic (non-screen trigger): the robot camera continuously runs person detection. A guest classified as APPROACHING or CLOSE triggers this function.
- Manual: the guest taps the **Start talking** button or any area of the idle screen.

Function description

- Actors: Robot (system), Guest.
- Purpose: start an interaction without the guest having to press anything.
- Interface: robot face (animated eyes and expressions), greeting subtitle, microphone indicator.

- Data processing: the detector estimates distance from the person's bounding-box height: CLOSE (< 1.5 m), APPROACHING (about 1.8–3 m), FAR (> 3 m). On approach the robot leaves Standby, moves RT-01 → RT-02, greets the guest and opens the microphone after 0.5 s if Auto-listen is on.

Function details

1. BR-GI-01: A running promotional workflow is stopped immediately when a guest approaches.
2. BR-GI-02: When the guest walks away (FAR or no person), the robot stops speaking and listening, saves (flushes) the chat session to the database, clears the room number and returns to RT-01.
3. BR-GI-03: The guest's voice also wakes the robot: any recognized speech during Standby closes the advertisement and moves to RT-03.
4. BR-GI-04: If the guest is silent for 45 seconds during a conversation, the session ends politely and the robot returns to RT-01.

Abnormal cases:

- Camera not available or permission denied → detection is disabled; the guest can still start by touch or voice.
- A session flush fails when the guest leaves → the error is logged and the robot still resets to RT-01.

3.3.2 Voice and Touch Conversation (Multilingual)

Function trigger

- Robot enters RT-03 after greeting, or the guest taps the microphone button.
- Guest changes language from the language selector in the header.

Function description

- Actors: Guest, Robot.
- Purpose: let the guest ask about facilities, amenities, policies, promotions and nearby attractions, and get recommendations, by voice or by touch.
- Interface: robot face, live transcript, AI answer as subtitles (can be hidden), audio wave, language menu (Tiếng Việt / English), Service menu, Stop speaking button.
- Data processing: speech is converted to text; the text, session ID, detected emotion and current room number are sent to `POST /api/v1/ai/chat`. The response returns the answer, detected language, current room number and flags such as `trigger_video_call`. The answer is spoken with TTS and shown as subtitles.

Function details

1. BR-CV-01: Supported languages are Vietnamese and English. The answer is returned in the language the guest spoke; the robot switches its listening language automatically.
2. BR-CV-02: Keywords open the matching screen directly: "menu", "order food", "room service" → Food Menu (3.3.4); "facilities", "amenities" → Facilities (3.3.3).
3. BR-CV-03: When the answer is a direction, the robot enters RT-05 Route Guidance, shows the route on the floor map and then a QR code the guest can scan to keep the directions on their phone. Saying "thank you" or "I remember" closes the guidance.
4. BR-CV-04: Conversation context (previous turns, room number) is kept for the whole session so follow-up questions are understood.

5. BR-CV-05: An empty transcript is ignored and the robot returns to RT-02.

Abnormal cases:

- The AI engine does not reply within 12 seconds → the processing lock is released and the robot returns to RT-02 so the guest can ask again.
- AI server unreachable → a fallback answer is used from the offline hotel knowledge (facilities, directions).
- Speech not recognized (noise) → no request is sent; the robot keeps listening.
- Guest taps **Stop speaking** → TTS stops immediately.

3.3.3 Hotel Facilities and Floor Map

Function trigger

- Service menu → **Facilities & Amenities**, or a voice question about facilities.

Function description

- Actors: Guest.
- Purpose: browse hotel facilities and see where they are.
- Interface: a modal of facility cards (Infinity pool, Restaurant, Spa, Fitness & Yoga, Rooftop Sky Bar, Reception, Parking & EV charging, Souvenir shop) with image, floor, opening hours and description; a **Show the way** action opens the floor map.
- Data processing: facility cards are read from the Amenity records (name, category, location, floor, open time, close time, description, active); upcoming hotel Events (title, location, start/end time) can be shown alongside. The floor map uses the saved map and Endpoints (waypoints).

Function details

1. BR-FC-01: Choosing a facility shows its location on the floor map with the route from the robot's position and a QR code.
2. BR-FC-02: The modal closes on Close, on tap outside, or by voice.

Abnormal cases:

- Map not loaded → the facility card is still shown; the map area displays "Map is unavailable".

3.3.4 Food Menu and Room Service Order

Function trigger

- Service menu → **Food menu**, a voice request to order food, or direct link `/food-menu`.

Function description

- Actors: Guest.
- Purpose: browse the menu and order food and drinks delivered to the room.

- Interface: category tabs (Appetizers, Soups & Porridge, Salads, …), item cards with image, price and portion, a cart with +/– quantity, room badge, order note, Review order and Place order.
- Data processing: the order is sent to `POST /api/v1/operations/room-service/orders` with room number, items (name, quantity, price, notes) and total amount. The server creates a RoomServiceOrder with one OrderItem per dish (item name, quantity, unit price, subtotal, note, link to the menu item and food item), links it to a SupportRequest, and notifies the Room Service/Kitchen department.

Function details

Field	Validation
Room number	Required before ordering. Pre-filled from the conversation if the guest already said it; editable via the room badge. Linked to the Room record when the number exists.
Quantity	Integer ≥ 1 per item; reducing to 0 removes the item.
Item note	Optional free text per item (e.g. no chili).
Order note	Optional free text for the whole order.
Unit price, Subtotal	Unit price is copied from the menu item at order time; subtotal = unit price × quantity.
Total amount	Σ subtotals, in VND.
Order number, Status, Progress	Generated by the server; status starts Pending, progress 0%.
Priority	NORMAL by default.

Business rules:

1. BR-FM-01: Only items that are available on the active menu are displayed (managed in 3.5.4).
2. BR-FM-02: Place order is disabled while the cart is empty or an order is being submitted.
3. BR-FM-03: On success the cart is cleared and a confirmation shows the order code, room, item count and total.

Abnormal cases:

- No room number → the room dialog opens with "Please enter the room number before ordering".
- Network/API error → the screen still confirms to the guest and the order must be re-entered by staff; this case is logged (to be replaced by a retry queue).
- The screen is used in portrait orientation → a "Please rotate the screen" guard is shown.

3.3.5 Submit a Guest Service Request

Function trigger

- The guest asks by voice, e.g. "Please send more towels to room 512", "I need a taxi to the airport", "The air conditioner is broken".

Function description

- Actors: Guest, Robot, AI Concierge Engine.
- Purpose: turn a spoken request into a service ticket for the right department without staff input.
- Interface: the robot confirms the request by voice and subtitles, including the ticket code.
- Data processing: the AI engine classifies the request (3.4.3), then creates a SupportRequest with ticket code, room number, description, department and status Pending, and sends Notifications to that department and to Reception.

Function details

Request type	Department	Ticket prefix
Food & beverage	Room Service / F&B	ORD-
Towels, cleaning, amenities	Housekeeping	HK-
Luggage, porter	Bell Services	BS-
Repairs (AC, lights, water)	Maintenance	MN-

Taxi, airport transfer	Taxi / Transportation	TX-
Talk to a person	Concierge	CCG-
Other	Reception	REC-

Business rules:

1. BR-SR-01: A room number is required for in-room services. If it is missing, the robot asks "Which room number should we send this to?" and creates the ticket after the guest answers.
2. BR-SR-02: The guest can ask about the status of an earlier request by room number; the robot reads the current status (Pending, In Progress, Completed, Cancelled).
3. BR-SR-03: Every ticket records its source as "HCRobot Concierge AI Chat" for reporting.

Abnormal cases:

- The guest never gives a room number → no ticket is created; the robot suggests contacting Reception.
- Database error → the robot apologizes and offers a live call with staff (3.3.6).

3.3.6 Request Human Support (Live Video Call)

Function trigger

- The guest says "I want to talk to a staff member" / "connect me to someone", or taps **Call staff** in the Service menu.

Function description

- Actors: Guest, Robot, Concierge staff (human-in-the-loop).
- Purpose: hand the guest over to a real person when the robot cannot help, including interpreting for international guests.
- Interface: a video call window (Messenger style) with call status (Calling, Ringing, Connected, Ended), timer, mute/camera toggles and End call.
- Data processing: the server creates a SupportRequest (CCG-xxxx, department Concierge) and a human support session (LIVE-xxxx), then sends an urgent notification to Concierge. Signaling runs over WebSocket and media over WebRTC peer-to-peer. When the call ends, the recording is uploaded to cloud storage and linked to the session.

Function details

1. BR-HS-01: The call is linked to the guest's room number if known, otherwise to "Main Lobby Kiosk".
2. BR-HS-02: The conversation transcript up to that point is attached so staff see the context before answering.

3. BR-HS-03: The idle timer and advertisements are paused during the call.

Abnormal cases:

- No staff answers → the call stays in Ringing; the guest can end it and the ticket remains Pending for follow-up.
- Camera/microphone permission denied → the call continues as audio-only or shows a permission error.
- Recording upload fails → the call record is kept without the video link and the error is logged.

3.3.7 Rate the Robot's Assistance

Function trigger

- Automatically after an interaction is completed, or via Service menu → **Rate service**.

Function description

- Actors: Guest.
- Purpose: collect guest satisfaction for the Admin statistics.
- Interface: a quick feedback modal with 1–5 stars and an 8-second countdown.
- Data processing: `POST /api/v1/ai/feedback` stores a Feedback record (rating, category, optional comment, guest name, room number, chat session).

Function details

1. BR-FB-01: Rating must be an integer from 1 to 5; default is 5.
2. BR-FB-02: If the guest does nothing for 8 seconds, the current rating is submitted automatically.
3. BR-FB-03: A feedback is submitted only once per modal opening.

Abnormal cases:

- Submission fails → the modal still thanks the guest and closes; the error is logged.

3.3.8 Idle Mode Advertisement Display

Function trigger

- Timing: the robot stays in RT-01 with no interaction for 60 seconds.
- Manual: Service menu → **Promotion videos & posters**, or Admin pushes an ad to the robot.

Function description

- Actors: Robot, Guest (viewer), Admin (content owner).
- Purpose: show announcements, promotions and hotel information while the robot is free.
- Interface: full-screen player for images and videos, with a progress bar.
- Data processing: the robot loads active advertisements and plays them in a loop ordered by priority.

Function details

1. BR-AD-01: Only advertisements with `is_active = true` are played, sorted by priority (1–10).
2. BR-AD-02: An image is shown for its `duration_seconds` (default 10 s); a video plays to its end.
3. BR-AD-03: Any touch, voice or detected guest closes the player immediately.
4. BR-AD-04: Ads never start during a conversation (RT-02 to RT-05) or while a modal is open.

Abnormal cases:

- No active ads → the robot stays on the idle face.
- A media file fails to load → it is skipped and the next ad is played.

3.4 AI Concierge Engine

This feature groups the non-screen server functions that make the robot understand, answer and act. They run inside the FastAPI backend and are called by the robot screen (3.3). Entities: ChatSession, ChatMessage, SupportRequest, Notification.

3.4.1 Speech Processing (Speech-to-Text and Text-to-Speech)

Function trigger

- STT: the robot sends recorded guest audio to `POST /api/v1/ai/stt/transcribe` when browser speech recognition is not available.
- TTS: every robot answer is converted to audio through `POST /api/v1/ai/tts`.

Function description

- Actors: Robot (system).
- Purpose: convert guest speech to text and robot answers to natural speech in Vietnamese or English.
- Interface: no screen; REST API.
- Data processing: STT tries online recognition first and falls back to the offline Faster-Whisper model (tiny, CPU, int8) with voice-activity filtering. TTS uses Microsoft Edge neural voices (vi-VN-HoaiMyNeural, en-US-JennyNeural) and caches generated audio in memory and on disk.

Function details

1. BR-SP-01: The voice is selected by the detected language of the answer.
2. BR-SP-02: Speech rate is configurable by Admin from 0.5× to 2.0× (default 1.0×) via voice settings.
3. BR-SP-03: Identical sentences reuse cached audio, so frequent answers play without regeneration.

Abnormal cases:

- Online STT fails → offline Whisper is used.
- Audio cannot be decoded → empty text is returned and the robot keeps listening.
- TTS provider unavailable → the client falls back to the browser's built-in speech synthesis.

3.4.2 Knowledge Retrieval and Answer Generation (RAG)

Function trigger

- Each guest question sent to `POST /api/v1/ai/chat` (or `/chat/stream` for streamed answers).

Function description

- Actors: Robot (system); Admin/Staff maintain the knowledge (3.6.4).
- Purpose: answer questions about facilities, amenities, policies, promotions and nearby attractions using only hotel-approved knowledge.
- Interface: no screen; REST API.
- Data processing: the request passes through a LangGraph pipeline: (1) Fast-path: answers matching an Admin-defined scenario (greetings, Wi-Fi password, opening hours) are returned from memory with no LLM call; (2) language detection; (3) intent routing; (4) retrieval of the most relevant passage from the ChromaDB knowledge base; (5) generation of a short concierge-style answer by the configured LLM (local Ollama, Gemini or Groq).

Function details

1. BR-RAG-01: Fast-path scenarios are checked before any LLM call; a scenario can also return an action (show directions to a destination, robot eye expression, start a video call).
2. BR-RAG-02: Answers are generated with a low token limit (about 70 tokens) so they are short enough to speak.
3. BR-RAG-03: Answers must be based on the retrieved hotel knowledge; the robot must not invent prices, policies or services.
4. BR-RAG-04: The answer is returned in the detected language of the question.

Abnormal cases:

- Primary LLM (Gemini) fails → automatic fallback to the local Ollama model.
- All LLMs unreachable or a 12-second timeout → an offline fallback answer is built from the hotel knowledge, or the message "Sorry, the connection to the AI server is temporarily unavailable".
- Corrupted vector index → the collection is rebuilt automatically from the knowledge vault.

3.4.3 Intent Extraction and Ticket Routing

Function trigger

- Runs on every chat turn after the fast-path check.

Function description

- Actors: Robot (system), Hotel Staff (receivers).
- Purpose: recognize when the guest wants a service and create the ticket automatically (see 3.3.5).
- Interface: no screen; internal service.
- Data processing: the intent router matches the question against keyword sets per department (Housekeeping, Room Service, Bell Services, Maintenance, Taxi, Concierge) and classifies it as service, FAQ, room-number answer or chit-chat. For a service intent, a finite-state step checks the room number, creates the SupportRequest in the background and sends Notifications to the target department and Reception.

Function details

1. BR-IE-01: The room number is extracted from the sentence (e.g. "room 512", "phòng 512") and remembered for the session.
2. BR-IE-02: When a service intent has no room number, the intent is kept as pending and the robot asks for the room; the ticket is created when the guest answers.
3. BR-IE-03: A Concierge intent creates a live support session and sets `trigger_video_call = true` so the robot opens the video call (3.3.6).
4. BR-IE-04: Ticket creation runs in the background so the spoken answer is not delayed.

Abnormal cases:

- Intent unclear → treated as FAQ and answered by RAG.
- Ticket insert fails → the error is logged; no notification is sent.

3.4.4 Conversation Session Management

Function trigger

- A new session ID is created each time the robot screen starts a conversation; it is closed when the guest leaves or the session times out.

Function description

- Actors: Robot (system).
- Purpose: keep conversation context and save conversation history for staff and admin review.
- Interface: no screen; APIs `POST /ai/session/reset`, `POST /ai/session/flush`, `GET /ai/sessions`, `GET /ai/sessions/{id}/messages`.
- Data processing: the session (chat history, room number, pending intent, last active time) is kept in server memory during the conversation, then flushed to the database as one ChatSession (room number, guest name, active flag) with its ChatMessages. Each ChatMessage stores the sender (guest or robot), text, language and the detected intent action.

Function details

1. BR-SS-01: Only the last 20 messages are kept in memory as context.
2. BR-SS-02: A session with no activity for 90 seconds is cleared from memory.
3. BR-SS-03: Empty sessions (no messages) are not saved.
4. BR-SS-04: Staff can press **Reset** on the robot to start a clean session immediately.

Abnormal cases:

- Server restarts during a conversation → in-memory context is lost; the next question starts a new session.
- Database unavailable at flush time → the flush fails and is logged; the conversation is not recoverable.

3.5 Staff Operations

This feature is the web/PWA workspace for hotel staff (`/staff/*`). Each staff account sees only its own department's work, except Managers who see all departments. Entities: SupportRequest, RoomServiceOrder, OrderItem, MenuItem, Notification, ChatSession, ChatMessage, Feedback, Department, Account.

All service requests follow one lifecycle:

Status	Progress	Set by
Pending	0%	Created by the robot, a guest or Reception; or released back by staff.
In Progress (food orders: Sent to Kitchen 25% → Cooking 50% → Ready 75% → Delivering 85%)	25–85%	Staff claims the request or advances its step.
Completed (Delivered)	100%	Staff marks the request done.
Cancelled / Declined	–	Staff or Manager cancels an invalid or duplicate request.

3.5.1 Department Dashboard

Function trigger

- Automatically after login (staff are routed to their department dashboard), or sidebar → **Dashboard**.

Function description

- Actors: Hotel Staff (Reception, Housekeeping, Bell Services, Maintenance, Room Service, Kitchen, Taxi, Concierge), Manager (Overview dashboard).
- Purpose: give each department a live view of its workload.
- Interface: KPI cards (Pending requests, In progress, Completed today, Staff on duty), a list or board of active tickets with room, guest, items, elapsed time and quick action buttons, plus a refresh button. The Kitchen dashboard shows orders as cards by cooking step; the Taxi dashboard shows pickup point, destination and vehicle; the Manager overview shows all departments together.
- Data processing: `GET /api/v1/operations/dashboard/{department}` loads counters and tickets; new tickets arrive in real time through the notification WebSocket.

Function details

1. BR-DB-01: A staff account only sees tickets of its own department; Manager/Admin sees all.
2. BR-DB-02: Counters refresh automatically when a ticket is created or its status changes.
3. BR-DB-03: Reception also receives a copy of every ticket created by the robot so the front desk is always informed.

Abnormal cases:

- API unavailable → the last loaded data is kept and an error toast "Cannot load data from server" is shown; the user can press Refresh.
- No tickets → empty state "No requests at the moment".

3.5.2 Manage Guest Requests

Function trigger

- Sidebar → **Requests** (`/staff/requests`), or click a new-request notification.

Function description

- Actors: Hotel Staff, Manager.
- Purpose: receive, claim, update and complete guest requests transferred by the robot or created at the desk.
- Interface: summary counters, status filter (All, Pending, In Progress, Completed), department filter (Manager only), search box, paginated list (10 per page) and a detail drawer with actions: **Claim**, next step, **Mark completed**, **Release**, **Cancel**, and **Create request**.
- Data processing: actions call `PATCH /api/v1/operations/tasks/{ticket_id}` with the new status, assignee and optional note. The server updates status, progress, assigned staff and account, and notifies the related parties.

Function details

Field	Description
Ticket code	ORD-, HK-, BS-, MN-, TX-, CCG-, REC- + number (read-only, unique).
Title / Description	What the guest asked for; staff may add a note.
Service type	Service type of the department (each type has a default priority).

Priority	NORMAL by default; can be raised by Manager/Admin.
Room / Location	Room number or area (e.g. Lobby), linked to the Room record when it exists.
Guest name	Default "Guest (Room xxx)" when created by the robot.
Items, Total amount	For orders: list of items and total in VND.
Source	From HCRobot, Front Desk or Guest App.
Status, Progress, Estimated completion	See the lifecycle table above.
Assigned staff / Assigned robot	Staff account who claimed the request, or the robot doing a delivery.
Created / Updated time	System time.

Business rules:

1. BR-RQ-01: Claiming a Pending request assigns it to the current staff and moves it to In Progress (food orders move to Sent to Kitchen).
2. BR-RQ-02: A staff member can hold at most 3 active requests at the same time.
3. BR-RQ-03: Moving a request back to Pending clears the assignee.
4. BR-RQ-04: Search matches ticket code, title, location and guest name; results are sorted with the newest and most urgent first.
5. BR-RQ-05: A request created manually by staff starts as In Progress if an assignee is chosen, otherwise Pending.

Abnormal cases:

- Claiming a 4th request → "You are handling 3/3 requests (#…). Please complete some before taking more."
- Update without any valid field → HTTP 400 "No valid update information".

- Ticket not found (deleted or wrong code) → HTTP 404 "Request #… not found".
- Two staff claim the same request at once → the last update wins and both lists refresh; the second staff sees the new assignee.

3.5.3 My Tasks

Function trigger

- Sidebar → **My Tasks** (`/staff/my-tasks`), or automatically after claiming a request.

Function description

- Actors: Hotel Staff.
- Purpose: focus on the requests the current staff member is handling.
- Interface: list of the user's active requests with progress bar, elapsed time, and step buttons (e.g. Cooking → Ready → Delivering → Delivered for food orders).
- Data processing: same update API as 3.5.2, filtered by the current account.

Function details

1. BR-MT-01: Only requests assigned to the current user with status In Progress (or a cooking/delivering step) are listed.
2. BR-MT-02: Completing a task moves it to History and frees one of the 3 task slots.

Abnormal cases:

- Request reassigned or cancelled by a Manager → it disappears from the list after refresh with an information toast.

3.5.4 Kitchen Menu Management

Function trigger

- Kitchen/F&B staff: sidebar → **Menu management** (`/staff/kitchen-menu`). The menu item is hidden for other departments.

Function description

- Actors: Kitchen staff, F&B staff, Admin.
- Purpose: maintain the food catalogue and the menus shown on the robot (3.3.4).
- Interface: three areas: Food library (master list of dishes), Menus (e.g. Breakfast, All-day dining, Drinks), and Menu items (dishes assigned to a menu with their selling price). Each has search, category filter, Add/Edit forms, availability toggle and Delete.
- Data processing: CRUD calls to `/api/v1/operations/kitchen/food-items`, `/menus` and `/menus/{id}/items`.

Function details

Field	Validation
Dish name (Food item)	Required.
Category	Required; default "Main course".
Description, image URL	Optional.
Preparation time	Minutes, integer > 0, default 15.
Available (Food item)	On/Off; default On.
Menu name	Required.
Menu category	Required (e.g. Food, Drinks).
Menu active	On/Off.
Price (menu item)	Number ≥ 0; currency VND.
Display order	Integer; lower numbers are shown first on the robot.
Available (menu item)	On/Off; default On.

Business rules:

1. BR-KM-01: A dish must exist in the Food library before it can be assigned to a menu; one dish can be in several menus with different prices.
2. BR-KM-02: Turning a dish or a menu off hides it from the robot immediately without deleting it.
3. BR-KM-03: Deleting a dish from the library removes it from all menus; past orders keep their own item name and price.

Abnormal cases:

- Empty dish or menu name → "Please enter the dish name" / "Please enter the menu name".
- Assign without selecting a menu or a dish → "Please select a menu first" / "Please select a dish from the library".
- Save fails → "Error saving, please try again"; form data is kept.

3.5.5 Conversation and Service History

Function trigger

- Sidebar → **History** (`/staff/history`).

Function description

- Actors: Hotel Staff, Manager.
- Purpose: review past robot conversations and guest service records to follow up and improve service.
- Interface: statistics (sessions, average rating, 5-star count), search box, paginated list of sessions (session ID, room, guest, time, rating, feedback) and a transcript viewer; Export button.
- Data processing: reads ChatSession, ChatMessage and Feedback via `GET /api/v1/operations/admin/conversations` and `/ai/sessions/{id}/messages`.

Function details

1. BR-HI-01: Search matches session ID, room, guest name, feedback text and transcript content.
2. BR-HI-02: History is read-only; nobody can edit a transcript.

Abnormal cases:

- No result → "No conversations found".

3.5.6 Notifications

Function trigger

- Real time: a new ticket, status change, robot alert or Manager directive. Staff open the bell icon or sidebar → **Notifications**.

Function description

- Actors: Hotel Staff, Manager, Admin.
- Purpose: alert staff immediately about requests or situations that need attention.

- Interface: bell badge with unread count, toast pop-up with sound, Notifications page with filters (All, Unread, Robot, Warning), search, mark read/unread, mark all as read, delete.
- Data processing: the browser keeps a WebSocket to `/api/v1/operations/ws/notifications?department=…`; history is loaded with `GET /operations/notifications`.

Function details

1. BR-NT-01: A notification is delivered only to the target department (and Reception for robot tickets).
2. BR-NT-02: Urgent and Warning types (e.g. live call request, robot fault) are highlighted in red and play an alert sound.
3. BR-NT-03: Clicking a notification marks it as read and opens the related request.

Abnormal cases:

- WebSocket disconnected → the client reconnects automatically every 5 seconds; missed notifications are loaded from the API on reconnect.
- Browser blocks audio → the visual toast is still shown.

3.5.7 Live Call Center and Recordings (Concierge)

Function trigger

- Concierge staff: dashboard tabs **Live call**, **Recordings**, **Taxi dispatch**, **Service feedback** (`/staff/concierge?tab=…`). An incoming call is announced by an urgent notification.

Function description

- Actors: Concierge staff, Manager.
- Purpose: provide personalized human assistance when the robot escalates (3.3.6), assist communication with international guests, dispatch taxis and review guest ratings.
- Interface: queue of incoming calls (room, guest, conversation so far), Accept/Decline, video call window, a recordings library with playback, a taxi dispatch board and a feedback list.
- Data processing: accepting sets the linked SupportRequest to In Progress and joins the WebRTC call; on end the recording is uploaded to cloud storage (`POST /operations/concierge/recordings/{session_id}`) and the human support session is updated with recording URL, duration, call start/end time, guest sentiment and wait time, then closed.

Function details

1. BR-LC-01: The staff member sees the robot transcript before accepting the call.
2. BR-LC-02: Every call is recorded and linked to its support session for quality review.
3. BR-LC-03: After the call, staff can create a follow-up ticket for another department.

Abnormal cases:

- Guest hangs up before acceptance → the call is marked missed and the ticket stays Pending.
- Connection drops during the call → both sides see "Call ended"; the partial recording is still uploaded.

- Recording file not available → playback shows "Recording unavailable".

3.6 Administration

This feature is the Admin Portal (`/admin/portal`), reachable only by accounts with the Admin role. Its tabs are Dashboard, Operations, Robot Control, Knowledge & AI, Robot Advertisements, Department, Account, Logs and Settings. Entities: Hotel, Floor, Room, Account, Department, Robot, Job, Endpoint, SupportRequest, ChatSession, ChatMessage, Feedback, Notification.

Common rule for every Admin function: any request without a valid Admin token is refused (HTTP 401/403) and the user is sent back to Login. Every create, update and delete is recorded in the audit log with the actor and the before/after values.

3.6.1 Admin Dashboard and Statistics

Function trigger

- Default tab after Admin login, or Admin Portal → **Dashboard**.

Function description

- Actors: Admin.
- Purpose: monitor guest interactions, service requests, guest satisfaction and robot status in one place.
- Interface: KPI cards (conversations today, open requests, completed requests, average rating, robots online), charts of requests by department and by status over time, rating distribution, recent activity feed and robot status cards.
- Data processing: `GET /api/v1/operations/admin/summary` and `GET /operations/analytics/summary` aggregate SupportRequest, ChatSession, Feedback and Robot data.

Function details

1. BR-AD-DB-01: Statistics can be filtered by time range (today, 7 days, 30 days).
2. BR-AD-DB-02: Average rating = sum of ratings ÷ number of feedbacks in the range, shown with one decimal.
3. BR-AD-DB-03: Figures refresh automatically when new tickets or feedback arrive.

Abnormal cases:

- No data in the selected range → cards show 0 and charts show "No data".
- Summary API fails → an error banner with a Retry button.

3.6.2 Operations Monitoring and Dispatch

Function trigger

- Admin Portal → **Operations**.

Function description

- Actors: Admin.
- Purpose: supervise all service requests across departments, intervene when needed, and dispatch new tasks to staff or robots.
- Interface: a table of all tickets (code, title, room, department, status, assignee, time) with filters and search; a detail panel to change status, reassign or add a note; a **Dispatch** form (title, room/location, department, priority, assigned staff or robot).
- Data processing: updates use `PATCH /operations/tasks/{id}`; dispatch uses `POST /operations/admin/dispatch` or `POST /operations/directives`. A directive (ManagementDirective) stores code, title, department, type, priority, location, description, status, assigned staff, ETA and creator; it creates the ticket and notifies the department.

Function details

1. BR-OP-01: Title and room/location are required to dispatch.
2. BR-OP-02: A dispatched task with an assignee starts In Progress; without an assignee it starts Pending.
3. BR-OP-03: Admin can change any ticket regardless of department and the 3-task limit.

Abnormal cases:

- Missing title or room → "Please enter title and room/location".
- Update or dispatch fails → "Error updating request" / "Error dispatching"; the form keeps its data.

3.6.3 Robot Control, Map and Navigation

Function trigger

- Admin Portal → **Robot Control** (sub-tabs LiDAR Map, Workflows, Camera), or Landing page → **LiDAR Map** (`/admin/map`).

Function description

- Actors: Admin, technician.
- Purpose: manage robot profiles and operating status, build and maintain the navigation map, define navigation endpoints and zones, and control the robot remotely.
- Interface: fleet panel (robot code, status, battery, location, online/last heartbeat); live 2D LiDAR map with the robot position; map tools (Connect LiDAR, Scan 360°, Save, Load, Reset, Lock map); endpoint (waypoint) and zone editors; teleop pad (forward, back, left, right, stop); **Navigate to** a selected endpoint; live camera view.
- Data processing: map and endpoint APIs under `/api/v1/map/*`; commands are forwarded to the robot over ROS 2 (Nav2 for navigation, SLAM Toolbox for mapping). Robot status comes from `GET /operations/fleet`.

Function details

Object	Main fields	Rules
Robot	Code, name, model, status (IDLE, NAVIGATING, INTERACTING, CHARGING, ERROR, OFFLINE), battery 0–100%, current location, online flag, IP, last heartbeat	Code is unique.
Endpoint	Name, type (dock, service, standby, room, elevator), floor, x, y, yaw, description, map	Name required; coordinates must lie inside the map.
Endpoint group	Name, description, member endpoints	Groups endpoints for patrol routes or wayfinding; an endpoint can belong to several groups.
Zone	Name, type (Keep-out, Speed zone, No-stop, Silent, Greeting, Service priority), floor, rectangle x, y, width, height, speed limit (m/s)	Robot never enters a Keep-out zone; speed is capped inside a speed zone.
Job	Code, title, robot, workflow, schedule, trigger (Manual, Schedule, Auto-detect, Guest tap, Voice), status (Pending, Running, Paused, Completed, Failed, Cancelled), current step / total steps, error message	One running job per robot.

Business rules:

1. BR-RC-01: When the map is locked, it cannot be reset or overwritten by a new scan.
2. BR-RC-02: Teleop commands are sent only while the button is held; releasing it sends Stop.
3. BR-RC-03: Maintenance and scheduled tasks are stored as Jobs and appear in the robot's activity history.

Abnormal cases:

- LiDAR not connected → "LiDAR is not connected"; map tools are disabled except Connect.
- Navigation goal unreachable or blocked → Nav2 aborts, the robot stops and an alert is raised.
- Robot offline (no heartbeat) → its card turns grey and commands are disabled.

3.6.4 Knowledge Base and AI Scenarios

Function trigger

- Admin Portal → **Knowledge & AI**. Hotel Staff with permission can also update response content (Update knowledge).

Function description

- Actors: Admin, authorized Hotel Staff.
- Purpose: manage hotel information, amenities, policies and guest-service content that the robot uses to answer (3.4.2), and tune frequent answers.
- Interface: knowledge statistics (documents, chunks, last sync); document list with upload, edit (Markdown editor), delete and **Sync**; scenario list (title, category, keywords, Vietnamese/English answer, action, priority, active) with Add/Edit/Delete and **Reload cache**; voice settings (speech rate).
- Data processing: documents go to `/api/v1/rag/*`; text is extracted, split into chunks and embedded into ChromaDB; Markdown files are also saved in the knowledge vault. Scenarios use `/api/v1/ai/scenarios` and are cached in memory for the fast-path.

Function details

Field	Validation
Document file	.md, .txt, .pdf, .docx or images (.jpg, .png, .webp).
Scenario title	Required, max 150 characters.
Category	FAQ, Greeting, Facility, Service, Chit-chat.
Match keywords / match type	At least one keyword; match type Keyword, Exact or Regex.
Response (Vietnamese)	Required. Response (English) optional.
Action	None, Trigger video call, Show map, Open menu (+ payload, e.g. destination).

Priority, Active	Integer; higher priority is checked first. Inactive scenarios are ignored.

Business rules:

1. BR-KB-01: A new or edited document is searchable by the robot right after sync, without restarting the server.
2. BR-KB-02: Deleting a document also removes its chunks from the vector store.
3. BR-KB-03: Saving a scenario reloads the fast-path cache so the change applies to the next question.

Abnormal cases:

- Unsupported file type → upload is rejected with "Unsupported file format".
- PDF with no extractable text → the file is stored but no chunks are created; a warning is shown.
- Scenario not found → HTTP 404 "Scenario '…' not found".

3.6.5 Robot Advertisement Management

Function trigger

- Admin Portal → **Robot Advertisements**.

Function description

- Actors: Admin.
- Purpose: manage announcements, promotions and hotel information displayed by the robot in idle mode (3.3.8).
- Interface: grid of ads (preview, title, type, duration, priority, active), Upload file or Add by URL forms, Edit, Delete, active toggle and **Play now on robots**.
- Data processing: media is uploaded to cloud storage; ad records are managed via `/api/v1/advertisements`; Play now pushes the ad to all robot screens in real time.

Function details

Field	Validation
Title	Required.
Description	Optional.

Media	Upload JPG, PNG, WEBP, MP4, WEBM or MOV, max 100 MB; or a media URL. Required.
Media type	Image or Video (detected from the file).
Thumbnail	Generated for videos.
Display location	Where the ad is shown (e.g. robot idle screen, lobby).
Display duration	Seconds, default 10 (images only).
Priority	1–10.
Active	On/Off.

Abnormal cases:

- No file or URL → "Please choose an image or video file" / "Please enter the media URL".
- No title → "Please enter the ad title".
- Upload fails or file over 100 MB → error message; nothing is saved.

3.6.6 Department Management

Function trigger

- Admin Portal → **Department**.

Function description

- Actors: Admin.
- Purpose: manage the hotel departments that receive guest requests.
- Interface: department cards (name, code, description, active staff count, active status) with Add, Edit and Deactivate.
- Data processing: `/api/v1/operations/departments` CRUD; the service types handled by each department are listed with `GET /operations/service-types`.

Function details

1. BR-DP-01: Name is required; code is optional but must be unique (e.g. HOUSEKEEPING).
2. BR-DP-02: Delete is a soft delete: the department is set inactive and keeps its history; inactive departments receive no new tickets.
3. BR-DP-03: Staff count shows only active accounts.

Abnormal cases:

- Empty name → "Please enter the department name".
- Duplicate code → HTTP 400 "Department code '…' already exists".
- Department not found → HTTP 404.

3.6.7 Account Management

Function trigger

- Admin Portal → **Account**.

Function description

- Actors: Admin.
- Purpose: manage user accounts and access permissions for staff, managers, admins and robot devices.
- Interface: searchable, filterable table (avatar, full name, username, staff code, role, department, location, shift status, active) with Add, Edit, Reset password and Deactivate.
- Data processing: `/api/v1/operations/accounts` CRUD. New passwords are hashed before storage.

Function details

Field	Validation
Full name	Required.
Username	Required, unique, stored in lower case.
Staff code	Required (short badge, e.g. MS).
Role	Admin, Manager, Staff or Robot; default Staff.

Department	Required for Staff; must be an existing department (DEP-HOUSEKEEPING, DEP-BELL, DEP-TAXI, DEP-MAINTENANCE, DEP-RECEPTION, DEP-CONCIERGE, DEP-ROOMSERVICE, DEP-KITCHEN).
Default dashboard	Derived from role and department.
Email, Phone	Optional; valid email format.
Shift	e.g. Morning (06:00–14:00).
Assigned floors	e.g. Floor 1–5.
Notification channels	e.g. Web dashboard, tablet alert.
Fallback agent	Yes/No; fallback agents receive requests auto-reassigned after the escalation time (see 3.6.9).
Initial password	Default 123456 if left empty; user should change it at first login.
Location, shift status, avatar	Optional; status Available, Busy, Off shift.
Current task count	System-maintained; used for the 3-task limit (BR-RQ-02).

Business rules:

1. BR-AC-01: The role decides access: Admin → Admin Portal; Manager → all staff dashboards; Staff → own department; Robot → robot screen only.

2. BR-AC-02: Delete is a soft delete; a deactivated account cannot log in (HTTP 403) but its history is kept.
3. BR-AC-03: An Admin cannot deactivate their own account.

Abnormal cases:

- Empty full name or username → "Please enter the staff full name" / "Please enter the username".
- Username already exists → HTTP 400 "Username '…' already exists. Please choose another."
- Department ID does not exist → HTTP 400 with the list of valid departments.

3.6.8 Logs and Audit Trail

Function trigger

- Admin Portal → **Logs**.

Function description

- Actors: Admin.
- Purpose: manage conversation logs, activity records and system reports; track who did what and when (the project's non-functional requirement on user tracking).
- Interface: statistics by level and category, filters (level, category, actor, time range, keyword), paginated log table, detail drawer, request trace view (all steps of one request by correlation ID), audit log tab and **Export CSV**.
- Data processing: `/api/v1/logs`, `/logs/statistics`, `/logs/trace/{correlation_id}`, `/logs/audit-logs`, `/logs/export`. Each HTTP request receives a correlation ID that is written to every related log entry.

Function details

Attribute	Values
Level	DEBUG, INFO, WARNING, ERROR, CRITICAL
Category	AI_VOICE, ROBOT, DISPATCH, AUDIT, SYSTEM
Actor type	ROBOT, STAFF, ADMIN, SYSTEM, AI, GUEST

Audit entry	Action (e.g. LOGIN, LOGOUT, CREATE, UPDATE, DELETE), resource, actor, before state, after state, time

Business rules:

1. BR-LG-01: Logs and audit entries are read-only; nobody can edit or delete them from the UI.
2. BR-LG-02: Export applies the current filters.

Abnormal cases:

- Trace ID not found → "No log found for this trace".
- Export with no matching rows → an empty file with headers only.

3.6.9 System Settings

Function trigger

- Admin Portal → **Settings** (sections General, Security, Notifications, Integrations, Advanced).

Function description

- Actors: Admin.
- Purpose: configure system settings and operational parameters.
- Interface: General (hotel profile, time zone, robot default language and operating mode); Security (admin profile, change password, session policy, sign out other devices, robot guest-verification method: room number + PIN, NFC keycard or face recognition); Notifications (hardware alerts such as low battery < 15% and connectivity loss, human hand-off escalation rules, channels); Integrations (AI provider, TTS provider, cloud storage); Advanced (data retention, anonymize guest voice data, export data archive, clear map cache, factory reset robot preferences).
- Data processing: settings are saved to the server and pushed to robots on their next sync.

Function details

1. BR-ST-01: Escalation default: alert the Manager when a request has no response after 5 minutes; auto-reassign after 3 minutes; broadcast to all on-duty staff after 8 minutes.
2. BR-ST-02: Danger-zone actions (factory reset, clear map cache) require a confirmation dialog.
3. BR-ST-03: **Reset settings** restores default values for the selected section only.

Abnormal cases:

- Invalid value (e.g. empty hotel name) → field error; nothing is saved.
- Save fails → "Cannot save settings"; previous values remain.

3.6.10 Robot Workflow Studio

Function trigger

- Admin Portal → **Robot Control → Workflows**, or the Studio view.

Function description

- Actors: Admin.
- Purpose: build robot behaviour scenarios without code, e.g. patrol the lobby, greet guests at the entrance, promote the spa at 6 pm.
- Interface: block-based builder with step categories Motion, Looks, Converse, Display, Voice, Sensing, Logic, Dispatch, Feedback; step types MOVE, GREET, COMMUNICATE, SHOW, SPEAK, LISTEN, RECOMMEND, CREATE_REQUEST, FEEDBACK; a kiosk simulator to preview; Save, Execute and Delete.
- Data processing: workflows (name, description, trigger type, steps, active) are stored via `/api/v1/workflows`. A workflow can be scheduled with a Schedule (time, repeat days or cron expression). **Execute** or a schedule creates a Job; each step is stored as a JobStep (type, parameters, status, result) and every state change as a JobEvent (type, message, severity).

Function details

1. BR-WF-01: A workflow needs a name and at least one step; a MOVE step must reference an existing Endpoint.
2. BR-WF-02: A running workflow is stopped immediately when a guest approaches (see BR-GI-01).
3. BR-WF-03: A scheduled workflow runs only when the robot is IDLE.

Abnormal cases:

- Referenced endpoint deleted → the step is marked invalid and the workflow cannot be executed.
- Step fails during execution → the Job status becomes Failed with an error message, and an alert is sent to Admin.

3.6.11 Hotel Content Management

Function trigger

- Admin Portal → **Hotel Content** (hotel content tab).

Function description

- Actors: Admin.
- Purpose: manage the hotel profile, floors, rooms, amenities and events that the robot presents to guests and uses to route requests.
- Interface: sub-tabs Hotel profile, Floors, Rooms, Amenities, Events, each with a searchable list and Add/Edit/Deactivate forms.
- Data processing: CRUD on the Hotel, Floor, Room, Amenity and Event entities; amenity and event changes are shown on the robot (3.3.3) at the next refresh.

Function details

Object	Main fields	Validation
Hotel	Name, address, phone, email, star rating, description, active	Name and address required; star rating 1–5.
Floor	Floor number, name, description	Floor number unique within the hotel.
Room	Room number, room type, floor, status, current guest name, QR code, PIN code, tag code, description	Room number unique within the hotel; status e.g. Available, Occupied, Cleaning, Out of order.
Amenity	Name, category, location, floor, open time, close time, description, active	Name, category, location, open and close time required.
Event	Title, location, start time, end time, description, active	End time must be after start time.

Business rules:

1. BR-HC-01: Deactivated amenities and events are hidden from the robot but kept for history.
2. BR-HC-02: A room's PIN code is used to verify the guest when the room-number + PIN verification method is enabled (3.6.9).
3. BR-HC-03: Requests and orders are linked to the Room record when the spoken room number matches an existing room.

Abnormal cases:

- Duplicate room or floor number → "Room/floor number already exists".
- Event end time before start time → "End time must be after start time".
- Deleting a floor that still has rooms → refused; the Admin must move or deactivate the rooms first.

3.7 Robot Monitoring

This feature groups the non-screen functions that run on the robot itself (Raspberry Pi 5, ROS 2) to report its health and keep people safe. Results are shown to Staff and Admin in 3.5.6 and 3.6.3. Entities: Robot, Job, Notification.

3.7.1 Telemetry and Heartbeat Reporting

Function trigger

- Timing: every 10 seconds while the robot software is running (ROS 2 telemetry node), plus on every status change.

Function description

- Actors: Robot (system); Admin and Staff consume the data.
- Purpose: let the hotel monitor each robot's operating status in real time.
- Interface: no screen; data appears on the Admin fleet panel and staff dashboards.
- Data processing: the robot sends a heartbeat to the backend health endpoint and publishes status, battery level, current location and pose. The server updates the Robot record (status, battery, location, `last_heartbeat`, `is_online`).

Function details

1. BR-TM-01: A robot is shown Offline when no heartbeat is received for 30 seconds (3 missed heartbeats).
2. BR-TM-02: A battery level below 15% raises a Low-battery warning to Admin and Maintenance.
3. BR-TM-03: Status values are IDLE, NAVIGATING, INTERACTING, CHARGING, ERROR, OFFLINE.

Abnormal cases:

- Backend unreachable → the robot logs "Cannot connect to server" and keeps retrying; local safety functions keep working.
- Network (Tailscale VPN) drops → the robot is marked Offline; a Connectivity-loss alert is sent.

3.7.2 Obstacle Safety and Abnormal-Condition Alerts

Function trigger

- Continuous: the safety watchdog runs every 50 ms whenever the robot can move (navigation, teleop or workflow).

Function description

- Actors: Robot (system); Admin and Maintenance receive alerts.
- Purpose: prevent collisions with guests and objects and alert staff when abnormal conditions occur.
- Interface: no screen; the robot face shows a warning expression, and alerts appear in Notifications and Logs.
- Data processing: every velocity command passes through the safety controller before reaching the motors. It checks the LiDAR front cone (about ±35°) and the four ultrasonic sensors (front, rear, left, right) and publishes a safety state SAFE, SLOWDOWN or EMERGENCY_STOP.

Function details

Condition	Threshold	Robot action
Obstacle in front (LiDAR)	≤ 0.80 m	SLOWDOWN: reduce speed.
Obstacle in front (LiDAR)	≤ 0.50 m	EMERGENCY_STOP: linear and angular speed set to 0.
Ultrasonic front / rear	≤ 65 cm	Block movement in that direction.
Ultrasonic left / right	≤ 25 cm	Block turning toward that side.
Robot inside a Keep-out or Slow zone	Zone polygon	Avoid the zone / cap speed (see 3.6.3).

Business rules:

1. BR-SF-01: Safety always overrides navigation, teleop and workflow commands.
2. BR-SF-02: An EMERGENCY_STOP longer than 30 seconds, a hardware error (LiDAR, motor driver, camera) or a navigation failure creates a Warning notification for Admin and Maintenance and a ROBOT log entry.
3. BR-SF-03: Movement resumes automatically once the path is clear and the state returns to SAFE.

Abnormal cases:

- LiDAR data missing → the robot treats the front as blocked and refuses to move forward.
- Sensor reading invalid (out of range) → the reading is ignored and the last valid value is used.

4. Non-Functional Requirements

4.1 External Interfaces

HCRobot communicates with four groups of external elements: users, robot hardware, third-party software services, and the network between the robot and the server.

User interfaces

Interface	Users	Platform	Requirement
Robot kiosk screen (/robot)	Guests	Touchscreen on the robot, landscape; installable PWA on phones (portrait layout)	Large touch targets (≥ 44 × 44 px), animated robot face, subtitles, no OS or logout controls visible.
Staff console (/staff/*)	Hotel staff, Manager	Desktop browser and mobile PWA	Responsive from 360 px width; sidebar on desktop, bottom navigation on mobile.
Admin portal (/admin/*)	Admin	Desktop browser (≥ 1280 px recommended)	Tab-based console with tables, charts, map and camera views.
Voice	Guests	Robot microphone and speaker	Vietnamese and English speech in and out.

All screens use Unicode (UTF-8) text and fonts that render full Vietnamese diacritics.

Hardware interfaces

Hardware	Connection	Purpose
Raspberry Pi 5 (Ubuntu + ROS 2)	Main robot computer	Runs ROS 2 nodes, kiosk browser, camera stream.
RPLIDAR 2D laser scanner	USB serial	Mapping (SLAM), localization, obstacle detection.
Camera	USB / CSI	Guest detection, live view, video calls.

ESP32 micro-controller	Serial (UART)	Motor PWM control, wheel encoders, IMU (MPU), ultrasonic sensors.
DC motors with encoders, differential drive	Via motor driver on ESP32	Robot movement and odometry.
4 ultrasonic sensors (front, rear, left, right)	ESP32 GPIO	Close-range obstacle safety.
Touchscreen, microphone, speaker	HDMI/USB	Guest interaction.

Software interfaces

System	Used for	Interface
PostgreSQL (local or Supabase cloud)	Main relational database	SQLAlchemy 2.0 async (asyncpg), Alembic migrations
ChromaDB	Vector store for the knowledge base	Python client, persistent local storage
Ollama (local LLM), Google Gemini, Groq	Answer generation and intent extraction	HTTP REST APIs; provider chosen by configuration
Microsoft Edge TTS (optional ElevenLabs, OpenAI TTS)	Text-to-speech	Python SDK / HTTPS
Google Speech Recognition, Faster-Whisper	Speech-to-text	HTTPS / local model

Cloudinary	Storage of call recordings and advertisement media	HTTPS API
ROS 2 (Nav2, SLAM Toolbox, rosbridge)	Navigation, mapping, robot control	ROS topics; WebSocket via rosbridge for the web map

Communication interfaces

Channel	Protocol	Used between
REST API (/api/v1/*)	HTTPS, JSON, JWT Bearer token	All clients ↔ backend
Real-time notifications	WebSocket (WSS)	Backend → staff/admin consoles and robot
Live video call	WebRTC (peer-to-peer media) with WebSocket signaling	Robot ↔ Concierge staff
Robot ↔ server network	Tailscale mesh VPN with fixed 100.x.y.z addresses	Raspberry Pi ↔ backend server
ROS 2 inter-node	DDS (CycloneDDS)	Robot ↔ mapping/visualization laptop
API documentation	OpenAPI 3 (Swagger UI at /docs)	Developers

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4.2 Quality Attributes

4.2.1 Usability

1. A first-time guest must be able to ask a question and hear an answer without any instruction: the robot greets automatically and opens the microphone; at least 90% of test guests complete a facility question unaided.
2. A guest can place a room-service order of 3 items in under 90 seconds using touch only.
3. Hotel staff must become productive with the Staff console after at most 30 minutes of training; an Admin (power user) after at most 2 hours, including map and knowledge-base management.
4. A staff member can claim and complete a request in no more than 3 clicks from the notification.
5. Every screen supports Vietnamese and English; the robot answers in the language the guest speaks.
6. All errors are shown in plain language next to the field or as a toast, never as raw technical messages.
7. Interfaces follow common web usability conventions (consistent navigation, visible system status, confirmation before destructive actions) and WCAG 2.1 level AA contrast for text.
8. Robot subtitles can be hidden or shown, and the speech rate can be adjusted from 0.5× to 2.0× for guests who need it.

4.2.2 Reliability

1. Availability: the backend and robot service must be available 99.5% of the time during hotel operating hours (24/7), excluding planned maintenance announced 24 hours in advance and done between 01:00 and 05:00.
2. Degraded mode: if the cloud LLM fails, the system falls back to the local model; if all AI is down, the robot still answers common questions from fast-path scenarios and offline hotel knowledge, and still accepts touch orders.
3. Robot safety functions (obstacle stop) run locally on the robot and must keep working when the network or the server is down.
4. MTBF: at least 200 hours of operation between failures that stop the robot from serving guests.
5. MTTR: a failed robot application must restart automatically (system service) within 2 minutes; a server failure must be repaired within 4 hours.
6. Accuracy: intent routing must send at least 90% of service requests to the correct department; ticket totals must match the sum of item prices exactly (VND, integer).
7. Data durability: no confirmed ticket, order or feedback may be lost; the database is backed up daily and kept for 30 days.
8. Defect rate: no more than 1 open critical defect and 5 significant defects per release; overall below 2 defects/KLOC at acceptance testing.

Defect categories:

Category	Definition	Example

Critical	Loss of data, a safety hazard, or a main module completely unusable	Robot does not stop for an obstacle; tickets not saved; nobody can log in
Significant	A function fails but a workaround exists	Video call fails but the ticket is still created; wrong department for a request
Minor	Cosmetic or wording issue, no effect on the result	Misaligned button, missing translation

4.2.3 Performance

Item	Target (average)	Maximum
Fast-path scenario answer (server)	< 50 ms	200 ms
Robot voice answer, end of guest speech → start of robot speech	< 3 s	6 s
Processing timeout before the robot recovers	–	12 s
REST API for CRUD screens	< 500 ms	2 s
Page load of Staff/Admin console (after first visit, cached PWA)	< 2 s	4 s
New ticket → notification shown on staff screen	< 2 s	5 s
Video call connection after staff accepts	< 5 s	10 s

Obstacle detection → robot stop command	< 100 ms	150 ms
Robot heartbeat interval	10 s	–

Throughput and capacity:

1. The backend must handle at least 50 requests per second and 200 concurrent WebSocket connections on a single server.
2. The system must support at least 5 robots, 100 staff accounts and 500 rooms per hotel.
3. The database must hold at least 1,000,000 chat messages and 100,000 tickets without a noticeable slowdown on paginated lists (10–20 rows per page).

Resource utilization:

1. On the Raspberry Pi 5, all robot software (ROS 2 nodes, kiosk browser, camera stream) must use at most 80% CPU and 6 GB RAM on average.
2. Uploaded advertisement media is limited to 100 MB per file; avatars to 5 MB.
3. TTS audio is cached so repeated answers do not consume network or CPU again.

4.2.4 Security

1. Every API except login and health check requires a valid JWT Bearer token; tokens expire after 7 days.
2. Role-based access control: Admin, Manager, Staff and Robot can only reach the routes and data of their role and department.
3. Passwords are never stored in plain text; only salted hashes are stored (target: upgrade to bcrypt/Argon2 before production).
4. All traffic between clients, server and robot uses HTTPS/WSS or the encrypted VPN.
5. Security headers are sent on every response (X-Content-Type-Options, X-Frame-Options DENY, X-XSS-Protection, Referrer-Policy).
6. API keys and database credentials are kept only in server environment variables, never in the frontend or the repository.
7. The robot kiosk hides all system controls; exiting kiosk mode requires a hidden gesture plus a password.
8. Guest voice data and recordings are used only for service and can be anonymized or deleted according to the retention setting.

4.2.5 Auditability (User Tracking)

1. The system records who performed each important action and when: login, logout, failed login, create/update/delete of accounts, departments, tickets, knowledge, ads and settings, and robot commands.
2. Each audit entry stores actor type, actor ID and name, action, resource, before and after state, timestamp and correlation ID.
3. Audit logs are read-only and kept for at least 1 year.

4.2.6 Maintainability

1. The backend follows a layered structure (API endpoints → services → repositories → models); database changes are made only through Alembic migrations.
2. Every push runs the CI pipeline: frontend unit tests (Vitest) and production build, backend unit tests (Pytest).
3. New knowledge, scenarios, menus, ads and workflows can be added by Admin through the UI without code changes or server restarts.
4. The LLM, TTS and database providers can be switched by configuration (environment variables) only.
5. All APIs are documented automatically in OpenAPI/Swagger.

4.2.7 Portability and Compatibility

1. Web/PWA clients support the latest two versions of Chrome, Edge, Safari (iOS 16+) and Chrome on Android 10+.
2. The backend runs on Windows, Linux or macOS with Python 3.11 and can be deployed with Docker Compose.
3. The robot software runs on Ubuntu 22.04/24.04 with ROS 2 Humble or Jazzy; simulation is supported in Gazebo.
4. Text is stored and displayed in Unicode (UTF-8) throughout, including database, APIs, logs and exports.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	A service request must be received and categorized before the system transfers it to hotel staff when human intervention is required.
BR-02	Hotel staff are responsible for handling transferred service requests and updating their handling status.
BR-03	Guests may track the status of requests they have submitted.
BR-04	Guests may request assistance from hotel staff when they need human support.
BR-05	Guests may rate the robot’s assistance after an interaction.
BR-06	Only authorized administrators may manage user accounts and access permissions.
BR-07	Hotel information, promotions, and knowledge content must be managed through the designated staff or administrative functions.
BR-08	The system must generate an alert when abnormal robot operating conditions are detected.
BR-09	The robot displays hotel announcements, promotions, and information when it is idle.
BR-10	Conversation history and guest service activities must be recorded for service monitoring and review.

5.2 Common Requirements

1. The system shall support Unicode text in its interfaces and stored information.
2. The system shall support multilingual guest interaction through voice and display interfaces.
3. The system shall maintain relevant conversation context during an active interaction.
4. The system shall record the user associated with important processing functions.
5. The system shall make the latest recorded request status available to guests and hotel staff.
6. The system shall record conversation history, guest service activities, and robot operating events.
7. The system shall enforce permissions for administrative and hotel staff management functions.
8. The system shall present understandable feedback for completed operations, invalid input, and unavailable **services.**

5.3 Application Messages List

#	Message code	Message Type	Context	Content
1	MSG01	In line	No matching information is found.	No matching information found.
2	MSG02	In red, under the text box	A required input field is empty.	This field is required.
3	MSG03	In red, under the text box	An input value is invalid.	Please enter a valid value.
4	MSG04	Toast message	The preferred language is changed.	Language updated successfully.
5	MSG05	Toast message	A service request is submitted successfully.	Service request submitted successfully.
6	MSG06	In line	A service request cannot be submitted.	Unable to submit the request. Please try again.
7	MSG07	In line	The requested service status cannot be found.	Request status is unavailable.
8	MSG08	Toast message	Hotel staff update a service request status.	Request status updated successfully.
9	MSG09	Toast message	The request is transferred to hotel staff.	Request transferred to hotel staff.
10	MSG10	Toast message	A guest rating is submitted.	Thank you for your feedback.
11	MSG11	In line	Voice input cannot be recognized.	Voice input not recognized. Please try again.
12	MSG12	In line	The conversational AI service is unavailable.	AI assistance is temporarily unavailable.
13	MSG13	Alert message	An abnormal robot operating condition is detected.	Robot operating alert. Please check the robot status.
14	MSG14	Toast message	Hotel information or response content is saved.	Content updated successfully.
15	MSG15	In line	A user attempts an unauthorized action.	You do not have permission to perform this action.

16	MSG16	Toast message	System configuration is saved.	Settings updated successfully.

5.4 Other Requirements

1. The project shall provide a guest interface, a robot application, an application for hotel staff, and a web application for administrators.

2. The project shall deliver User Requirements, Software Requirement Specification, Architecture Design, Detail Design, Testing Document, Installation Guide, User Guide, and source code.

3. The completed system shall be developed, integrated, and tested against its documented functional requirements.

4. The system shall integrate its robot device and AI processing services to support the interactions defined in the system context diagram.